

1. Backdoor Devices

- Backdoor devices refer to hardware or software that allows unauthorized remote access bypassing normal authentication.
 - Can be intentionally installed by manufacturers, developers, or inserted by attackers.
 - Examples include hardware implants, rootkits, or hidden administrative accounts.
 - Used for persistent control, espionage, or data theft.
 - Detection is difficult; prevention involves hardware audits, firmware checks, and strong security policies.
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2. Distributed Denial of Service (DDoS) Attacks

- DDoS attacks overwhelm a target system with traffic from multiple sources.
- The goal is to make services unavailable by saturating network bandwidth or server resources.
- Common sources are botnets: networks of compromised devices controlled by attackers.
- Types include:
 - Volume-based attacks (flooding traffic).

- Protocol attacks (exploiting protocol weaknesses).
 - Application layer attacks (targeting web services).
 - Defenses: traffic filtering, rate limiting, scrubbing centers, firewalls, IDS/IPS, and cloud-based mitigation.
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3. Biometric Spoofing

- Biometric spoofing involves tricking biometric systems (fingerprint, face recognition, iris scan) with fake samples.
 - Attackers use fake fingerprints, photos, videos, or synthetic biometric data.
 - Spoofing undermines biometric authentication security.
 - Countermeasures include liveness detection, multi-factor authentication, challenge-response techniques, and improved sensor technology.
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4. Linux Hacking

- Linux is a popular OS for servers, IoT devices, and desktops, often targeted due to its open-source nature.
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Common Linux hacking vectors:

- Exploiting unpatched vulnerabilities.
 - Weak SSH configurations.
 - Privilege escalation via sudo or kernel exploits.
 - Exploiting misconfigured permissions and services.
 - Tools used include Metasploit, John the Ripper, Hydra, and custom scripts.
 - Defenses: regular updates, strong passwords, firewalls (iptables), SELinux/AppArmor, and auditing.
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5. Linux Backdoors

- Backdoors in Linux systems are hidden programs or scripts allowing attackers persistent access.
- Examples include:
 - Modified binaries.
 - Malicious kernel modules.

- Hidden user accounts.
 - Cron jobs running malicious scripts.
 - Detection via integrity checks (Tripwire), log monitoring, and rootkit scanners (Chkrootkit, Rootkit Hunter).
 - Prevention includes least privilege, patching, and monitoring.
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6. Intrusion Detection Systems (IDS), Honeypots, and Firewalls

- **IDS:** Monitor network or system activities to detect malicious actions or policy violations.
 - Types: Network-based IDS (NIDS) and Host-based IDS (HIDS).
 - Techniques: Signature-based (pattern matching), anomaly-based (behavioral).
- **Honeypots:** Decoy systems designed to attract attackers, gather intelligence, and distract from real assets.
 - Can be low-interaction (limited services) or high-interaction (fully functional).
- **Firewalls:** Devices or software that control incoming and outgoing network traffic based on

rules.

- Types: Packet-filtering, stateful inspection, proxy firewalls.
- Firewalls help block unauthorized access but do not detect attacks once inside (unlike IDS).